

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			Indicative content: Method - Put the 200 cubes into a tub. - Shake the tub and gently throw the cubes into a tray or a table. - Record the number of cubes that have landed with the shaded face upwards and remove from the tray. - Put the remaining cubes back into the tub. - Repeat steps 2 and 3 several times. Analysis - Plot a graph of the number of radioactive atoms remaining against the number of throws. - Draw a curve of best fit. - Draw a line across from 100 cubes and read off the number of throws – this is the half-life. 5–6 marks Comprehensively accounts for the method and analysis. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3–4 marks Comprehensively accounts for the method or the analysis OR gives a partly correct account of each. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Gives a limited description of the method or the analysis. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>	6			6		6

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Second graph	1			1		
	(c)	(i)		$\frac{270}{6}$ or 45		1		1	1	1
		(ii)		45 (1) 83 (1)		2		2	2	2
		(iii)		Reference to a VERY large number of dice / nuclei OR reduces anomalies or similar			1	1		1
				Question 8 total	7	3	1	11	3	10